

TIRATH BHATHAWALA

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Education

Universiteit van Amsterdam

Master of Science in Artificial Intelligence

Sept 2026 – Jun 2028

Amsterdam, Netherlands

SVKM's Dwarkadas J. Sanghvi College of Engineering

B.Tech in Computer Engineering with Honors in Intelligent Computing (CGPA: 9.32)

Nov 2022 – Jun 2026

Mumbai, India

Experience

BrandContext.ai

AI Intern

Mar 2025 – Aug 2025

Hybrid, Mumbai, India

- Developed an agentic AI system powered by dynamic graphs and **real-time factory operation databases** to support supervisor workflows, **detect anomalies**, and dynamically incorporate worker-input special conditions.
- Evaluated and productionized computer-vision pipelines for retail supermarkets, fine-tuning **YOLOv11** multi-class detectors for Xiaomi (**0.87 mAP@50** across **over 500 stores** in India).
- Built legally-compliant **speaker-diarization** pipelines for sales-customer calls using open-source (**SpeechBrain**) and commercial (**ElevenLabs**) stacks to enable accurate speaker labeling and downstream analytics.

Indian Institute of Technology, Patna

Research Intern

Oct 2024 – May 2025

Remote, Patna, India

- Developed a heuristic system to dynamically **optimize academic peer review lengths** by calculating composite scores based on **information density**, argument strength, and readability metrics. **Published in Scientometrics**. 📄
- Developed a novel LLM-argumentation system for peer review, evaluating 900+ ICLR reviews using leading LLMs (**Llama 3.1, Mixtral, Gemini 2.0, GPT-4o**), achieving high inter-annotator agreement (**Cohen's Kappa = 0.834**).

UIUC+ Summer Research Program

Research Assistant

Jun 2024 – Jul 2024

Remote, Urbana-Champaign, USA

- Investigated **C-to-Rust transpilation**, evaluating 5+ papers, transpilers, and CodeLLMs for conversion reliability.
- Analyzed **LLM-generated C/Rust code**, categorizing bugs to improve code-quality evaluation frameworks.

Projects & Publications

AntiBug: Runtime-Aware Multimodal Agentic Program Repair | *LangGraph, Next.js, TypeScript*



- Built a **12-agent LangGraph pipeline** for runtime-aware program repair, integrating a **Next.js telemetry SDK**, stack traces, user-event context, and DOM screenshots for root-cause diagnosis and **automated patch synthesis**.
- Evaluated on **69 real-world bugs** from Cal.com and Appsmith, fixing **32/69 issues vs 9/69 for GPT-4o**.
- Achieved **72.5% preference across 549 blind A/B patch evaluations** by professional developers.
- Publication:** Under Review at *Automated Software Engineering Journal (Special Issue)*.

FRAC-MAS: Conformal Multi-Agent Fracture Radiograph Analysis | *Grad-CAM, PyTorch*



- Engineered a **multi-agent clinical decision support system** that integrates a 4-model vision ensemble (MaxViT, RAD-DINO, YOLO, DenseNet) with specialized LLM agents and delivers **verifiable, RAG-grounded reasoning**.
- Implemented **conformal prediction** for statistically grounded differential diagnoses, achieving **92.0% empirical coverage**; stacking achieved a **96.4% detection rate** on an external Roboflow dataset.
- Publication:** Accepted to *The 2nd Agentic AI for Medicine Workshop at MICCAI 2026*.

DigniFy: Multi-Modal Multilingual Hate Speech Detection | *HuggingFace, TensorFlow, FastAPI*



- Engineered an AI Agent integrating modality-specific models, enabling **unified hate speech detection** through interactive orchestration across four LLMs (**DeepSeek R1, GPT-4o, Gemini 2.0, LLaMA 4.0**).
- Fine-tuned **DistilRoBERTa** for text analysis, implemented OCR for image extraction, and constructed an audio processing pipeline with STT-TTS, merging all modalities into a comprehensive video processing architecture.
- Publication:** Accepted to *IEEE CONECCT 2026 (12th Edition)*.

Technical Skills

Languages: Python, C/C++, Java, Rust, SQL, JavaScript, HTML/CSS.

AI/ML Frameworks: PyTorch, TensorFlow, Transformers, HuggingFace, scikit-learn, OpenCV, LangGraph, LangSmith.

Tools/Cloud: Linux, Git, Docker, Azure AI, FastAPI, Playwright.

Relevant Coursework: AI, ML, DL, NLP, Data Structures & Algorithms, Data Warehousing & Mining, Advanced DBMS.

Achievements

- National AI Examination (GATE DS&AI):** Achieved All India Rank 499 (Top 0.8%) out of 60,000+ candidates.
- Amazon ML Challenge 2024:** Secured Rank 292 out of 70,000+ participating teams nationwide.
- CodeBounty 2024:** 1st Place Winner in a college competitive programming contest out of 100+ registrations.